

Jul 12, 2021

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

In the context of cryptography, (a) describe the Kerckhoffs principle, (b) explain the purpose of key management, (c) explain the notion of symmetric key cryptography, and (d) explain the notion of asymmetric (or public) key cryptography.

Question 2 [10 points]

Define the notions of Confidentiality, Integrity, and Availability.

Part 2

Question 3 [40 points]

In the context of the SAML standard, (a) explain its main goal, (b) describe the two scenarios for the Web Single Sign On profile, namely Identity provider initiated and Service provider initiated, (c) explain how is it possible to guarantee trust in the assertions consumed by the relying party (hint: remember that assertions are cryptographically signed), and (d) explain how it is possible to avoid man-in-the-middle attacks.

Question 4 [10 points]

Briefly describe the notion of Data Protection Impact Assessment (DPIA) according to the Art. 35 of the GDPR.

Please, remember...

On each sheet, make sure to write: Name Surname Student ID number Question number	The file containing the scan of your answers should be named as follows: surname-name-studentIDnumber.pdf
---	---